

LOCALIZATION IN THE COMMUTANT, STAGE 16:

the commutant tower, operator growth, and transport memory.

What the companion notes establish, what the network forces,
and where an estimator is actually free

10 October 2026

Abstract

This stage starts from two companion notes and from the experimental session’s tests of stage 13, all supplied by the user. The notes are *Branching, angular geometry and joint dynamics* and *Wick geometry, covariance holonomy and the closure of Kikuchi dynamics*.

- We audit the notes’ claims numerically, all of which hold.
- We prove what in them is forced for the network, and we measure what decides between design directions.
- We test two non-standard estimator designs that the notes suggest; both fail, and we say exactly why.
- *The commutant tower (proved; checked).* The weight ensemble’s symmetry $O(n)$ has the Brauer algebra (pairings) as its commutant on tensors. The ReLU fold’s symmetry B_n has even-block partitions, and hidden-unit relabelling S_n has all partitions. For 4 and 6 legs the dimensions are $3 < 4 < 15$ and $15 < 31 < 203$.
 - Annealed pre-activation statistics are pairings: Gaussian plus dilation, exactly the companion’s mixture formula $\kappa_4 = \text{Cov}(Q)$. In particular annealed odd cumulants of pre-activations vanish.
 - Folds create the other blocks. The “bulk three-site structure” left in v56 is the quenched three-block sector.
 - The hard-core normalizer theorem says that the hyperoctahedral group, $B_n = \text{Aut}(\mathbb{Z}^n) \cap O(n)$, is the only transport keeping neuron-diagonal data closed. Gaussian weights are almost surely not in it, so the slices must be regenerated from transported legs: *transport memory* is forced.
- *Operator growth (measured on official network 0).* As functions of the input, the neurons’ first-chaos share falls from 73% (layer 1) to 19% (layer 16). The mean degree of their higher chaos grows linearly, $2.8 \rightarrow 17.1$, about one degree per layer. A fixed-basis (Pauli-path, input-Hermite) expansion is therefore hopeless at depth. The layer-wise Wick re-centring of Ebrahimi-Fard–Patras–Tapia–Zambotti, a frame that co-evolves with the instance’s weights, is what makes the problem low-order. The interdependence of representation and weights is not optional; it is forced.
- *Where the correction lives (measured).*
 - 60–66% of what any estimator must add to the Gaussian closure is an isotropic function of each row’s field (the Brauer and dilation sector). The rest is anisotropic.
 - The exact first-order chain captures 99.1–99.7% of the correction.
 - Its residual is incoherent and born late, consistent with the experimental session’s finding that 0.4–3.7% of v56’s error lies along the mean.

- *Two designs tested and ruled out.*
 - The leaf (wall-sum) estimator, using the companion’s Dirac energies (carré du champ), fails by a factor 10^5 already at layer 2. The global Dirac energy exceeds the transverse one by the Poincaré deficit, the higher-chaos energy, and the exact identity relies on cancellations between layers.
 - Minimal-sensitivity Wick frames do not beat the centred (BPHZ) frame.
- *What is free.* We state a row-exchangeable self-calibration theorem: an estimator can learn its own correction from its own weights. We then give the design space that remains: frame family, diagram blocks, rows, layers, instruments.

Checks: `scripts/verify_stage16.py`.

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1 The picture, and an accounting of stage 15

1.1 An accounting of stage 15

Stage 15 refused to assume that an estimator is a law propagator, and derived the error cocycle from first principles. Its proposed “theory-native estimator” was still chain-shaped, however. It kept an exact backbone, transported births and allocated effort; the Heisenberg reads and the knapsack are reorganisations of the same object. This stage asks the prior question: which features of a chain are *forced* by the mathematics, and which are free? Every claim below is labelled forced (a theorem or a measurement that leaves no choice) or free (a design freedom), and the free ones are tested where possible.

1.2 What is forced

1. **Co-evolving frames** (Section 4; measured). Network 0’s neurons, as functions of the input, have mean higher-chaos degree 2.8 at layer 1 and 17.1 at layer 16. A representation in a fixed basis is hopeless. The representation must be re-centred at every layer by the instance’s own law, the Wick map $\mu_l^{-1} \star \text{id}$. The estimator’s frame therefore depends on the weights dynamically.
2. **Algebraic sectors** (Section 3; proved). Annealed pre-activation statistics are Brauer elements (pairings), which covers the Gaussian and the dilation. The folds create every other partition block. The instance-specific (quenched) parts are orthogonal to all annealed ones.
3. **Transport memory** (Section 5; proved). No closure on neuron-diagonal data alone can be exact under a transport outside the hyperoctahedral group $B_n = \text{Aut}(\mathbb{Z}^n) \cap O(n)$. Gaussian weights are almost surely outside it. Births must be carried with their own transported legs, or recomputed.
4. **Incoherence of the residual** (stage 15; confirmed by the experimental session). Once the modular sector is right, the error is a sum of independent row-births, and no global correction reaches it.

1.3 What is free, and what testing it showed

freedom	candidate	status
frame parameters	minimal-sensitivity (PMS) Gaussian frames	no gain over centring (checked)
frame family	non-Gaussian (mixture, skew) frames	open
decomposition of the mean	leaf/wall sum with Dirac energies	fails without transverse conditioning (checked)
which blocks to evaluate	partition-algebra diagram selection by annealed variance	framework; open
which rows	row-exchangeable self-calibration	theorem; protocol for test
which layers	cocycle allocation (late layers)	measured value; untested in a system
instruments	task-preserving branching on collective walls	theory; cost doubles per branch

1.4 The numbers that decide

quantity (official network 0)	value
first-chaos share of neurons, layer 1 $\rightarrow$ 16	0.73 $\rightarrow$ 0.19
mean degree of higher chaos, layer 1 $\rightarrow$ 16	2.8 $\rightarrow$ 17.1 (about +1 per layer)
isotropic (field) share of truth – Gaussian closure, layers 4–16	0.60–0.66
share of truth – Gaussian closure captured by the exact first-order chain	0.991–0.997
wall-sum estimator with global Dirac energies, layer 2	MSE 3.5×10^{-2} (Gaussian closure 1.5×10^{-7})
Dirac energy / variance of pre-activations, layers 1, 2, 4, 16	1.00, 1.47, 2.54, 13.7

2 Audit of the companion notes

2.1 Branching, angular geometry and joint dynamics

The note reads Zhou’s paper constructively. We checked its finite statements and kept the following.

- *Poisson tensor growth.* $\kappa_t = e^{t(\chi_V - d)}$ is a Poisson mixture of normalized characters of $V^{\otimes k}$. The label chain $K(\lambda, \nu) = c_{\lambda\nu}^\nu \dim V_\nu / (d \dim V_\lambda)$ is stochastic. For spin $\frac{1}{2}$ appended to spin j the branches are $p_\pm = (j+1)/(2j+1), j/(2j+1)$ (checked at $j = \frac{1}{2}, 1, \frac{7}{2}$).
- *Multiplicity as a logical qubit.* In the three-spin decoherence-free subsystem with the stated basis, $S_{12} = -Z$, $S_{23} = \frac{1}{2}Z + \frac{\sqrt{3}}{2}X$ and $[S_{12}, S_{23}] = -i\sqrt{3}Y$ (checked exactly). Appending a spin preserves the multiplicity register even conditionally on the branch, because each branch effect is scalar on V_j (Schur).
- *The angular tower.* $|\psi_\theta\rangle = (|0\rangle + \theta)/\sqrt{2}$ gives $\text{Tr}(R_{W,m}A_{W,m}) = \int f_W d\sigma$ and $T_m^*B_{m-1} = B_m$. These are exact, including for coupled registers. The isotropic reference is $G_1 = \text{diag}(\frac{1}{2}, \frac{1}{2n}, \dots)$

(checked: 0.5, 0.0998–0.1004 at $n = 5$), faithful and non-tracial, with modular phases n^{iu} on $|0\rangle\langle i|$.

- *The correction to Zhou v1, Theorem 4.3.* For the signed Poisson walk, $\mathbb{E}[Z_a^2 Z_b^2] = ab + 2a^2 + a$. Monte Carlo gives 3.021 against 3.010; the product of separate means would be 1.33. The multi-time formula must be read as nested conditional expectations. Stage 15 cited that theorem only for the restriction mechanism, which stands.
- *The product-defect identity* $P_t(AB) - P_t A P_t B = \int_0^t P_s \Gamma(P_{t-s} A, P_{t-s} B) ds$, the Dirac realization $\| [D_G, \pi(A)] \Omega_\rho \|^2 = \text{Tr}(G C_\rho(A))$, moving contacts, and protected logical observables. We use the first two in Section 6.

What does not transfer by itself:

- The angular representation needs the cell integrals, so it is exact but not cheap.
- Growth needs a real increment.
- The note says both of these itself.

2.2 Wick geometry, covariance holonomy and Kikuchi closure

- *Holonomy.* For the symmetric-squeeze connection $\dot{T} = \frac{1}{2} \dot{Q} Q^{-1} T$ around a contact-preserving square of side a in $Q = \begin{pmatrix} 1 & v \\ v & 1+u \end{pmatrix}$, the transport is exactly orthogonal ($|TT^\top - I| \leq 5 \times 10^{-13}$). Its angle over $a^2/4$ is 0.985, 0.965, 0.935 at $a = 0.02, 0.05, 0.1$, so the leading coefficient $\frac{1}{4}$ is confirmed with an $O(a^3)$ correction.
- *Hard-core normalizer.* The coefficient of z_i^2 in $(Rz)_j (Rz)_k$ is exactly 0 for a signed permutation and 0.39 for a random rotation.
- *Mixture fourth cumulant.* $\kappa_4(X) = \text{Cov}(Q_{ij}, Q_{kl}) + \text{Cov}(Q_{ik}, Q_{jl}) + \text{Cov}(Q_{il}, Q_{jk})$. Monte Carlo gives 1.573, 0.538, -0.007 against 1.558, 0.518, -0.001 .
- *Realization relations.* For $B \sim \text{Bern}(p)$, $b^2 - b = 0$ but $:b^2: - :b: = -2p(B - p)$ (checked). Our gates satisfy $g^2 = g$, so a Wick deformation must respect this relation. That is one reason the centred Gaussian frame works on pre-activations, where it is nowhere violated, and not on gates.
- *The all-degree sparsifier and the Poisson-coupled weights.* They concern a mean-estimation dynamics (sampling), so they are bounded by stage 9's sampling law; we do not use them as estimators. The convex variational principle with equalized edge energies is the exact form of stage 15's knapsack (Section 7).

3 The commutant tower: Brauer, even partitions, partitions

3.1 Three symmetry groups and their tensor commutants

Three groups act on a hidden layer's $\mathbb{R}^n$.

- $O(n)$, under which the Gaussian weight ensemble is invariant on each side.
- The hyperoctahedral group B_n of signed permutations, which commutes with the fold $|\cdot|$.

- S_n , the relabelling of hidden units, which commutes with relu .

Theorem 3.1 (the commutant tower; classical, checked). *For $n \geq 2k$, the algebras of operators on $(\mathbb{R}^n)^{\otimes k}$ commuting with the diagonal action are as follows.*

- $O(n)$: the Brauer algebra $\text{Br}_k(n)$, spanned by perfect matchings of the $2k$ legs.
- B_n : the span of set partitions of the $2k$ legs with all blocks of even size.
- S_n : the partition algebra $\text{P}_k(n)$, spanned by all set partitions.

So $\text{Br}_k \subset \text{P}_k^{\text{even}} \subset \text{P}_k$.

Proof. An operator commuting with all coordinate permutations has entries that depend only on the equality pattern of its $2k$ indices, a set partition. Commuting also with the sign flips forces every block to contain an even number of legs. The $O(n)$ case is Brauer’s theorem. $\square$

Check. Invariant dimensions are $\frac{1}{|G|} \sum_g \chi(g)^{2k}$ for the finite groups and $\mathbb{E}(\text{Tr } O)^{2k}$ for Haar $O(n)$.

- $2k = 4$: S_4 gives 15, B_4 gives 4, and $O(6)$ gives 3.01 ± 0.02 . The counts are 15 (Bell), 4 (even partitions) and 3 (pairings).
- $2k = 6$: 203, 31, and 15.17 ± 0.22 ; the counts are 203, 31, 15.

3.2 Which statistics live where

Proposition 3.2 (annealed statistics; proved). *Let W_l have i.i.d. $N(0, \sigma^2)$ entries independent of the upstream state, and $z_l = W_l h_{l-1}$.*

1. $\mathbb{E}_{W_l}[\kappa_k(z_l)] = \sigma^k \sum_{\pi} \prod_{(a,b) \in \pi} \delta_{i_a i_b} \cdot \langle \kappa_k(h_{l-1}), \pi \rangle$, summed over perfect matchings π of the k legs. Annealed pre-activation cumulants are Brauer elements; in particular all annealed odd cumulants of pre-activations vanish.
2. Post-activation annealed statistics are S_n -invariant, hence partition-algebra elements. Odd blocks appear because relu breaks the sign symmetry. Even non-pairing blocks such as δ_{iiii} come from the fold and odd blocks such as δ_{iii} from the asymmetry.
3. If $z \mid Q \sim N(0, Q)$, then $\kappa_4 = \text{Cov}(Q_{ij}, Q_{kl}) + \text{two permutations}$, a pure pairing element. With $Q = qI$ (the dilation) it is $\text{Var}(q)(\delta\delta + \delta\delta + \delta\delta)$. Upper-geometry fluctuations create only Brauer-type cumulants. The dilation package of stage 13 is the Brauer κ_4 .

Proof. (1) Gaussian Wick calculus for $E[W_{i_1 j_1} \cdots W_{i_k j_k}]$ pairs the factors, identifying row and column indices together. (2) Relabelling of hidden units is a symmetry of the ensemble of networks, while sign flips are not. (3) Condition on Q and subtract the three pair products (checked in Section 2). $\square$

Corollary 3.3 (where the residual lives). *Stage 15’s isotropic sector is the Brauer span. Three-block content of pre-activations, the “stars”, has zero annealed part, so it is purely quenched: no counterterm, mean-field tier or annealed correction can reach it. Only contractions with the instance’s transported legs can. This is the experimental session’s “bulk three-site structure”. The dilation, by contrast, is Brauer, and the fitted counterterms had already absorbed it; that is why its exact package did not pay on v56.*

3.3 The noiseless subsystem of the ensemble

By Schur–Weyl–Brauer duality, $(\mathbb{R}^n)^{\otimes k} = \bigoplus_{\lambda} V_{\lambda}^{O(n)} \otimes M_{\lambda}$, where Br_k acts on the multiplicity spaces M_{λ} . The ensemble average over $O(n)$ is the twirl that keeps only the multiplicity (leg-pattern, replica-overlap) information. This is the companion’s “information behind the label” with the roles reversed. The collective noise here is the randomness of the weights; it *decoheres the spatial factor*, where the instance-specific information lives, and preserves the multiplicity factor, which is the annealed (NNGP-type) theory. Annealed theory is the dynamics of the noiseless subsystem. Estimation of one instance is the recovery of what that twirl destroys, and a quantum Schur transform would not help with it: it gives access to multiplicities, which are classically cheap here.

4 Operator growth and the necessity of co-evolving frames

4.1 The chaos ladder of a real network

Regard a neuron $h(x)$ as a function of the Gaussian input and expand it in Wiener chaos, $h = \sum_k h_k$. Three energies are accessible by forward-mode differentiation alone:

$$F = |\mathbb{E}\nabla h|^2 = \|h_1\|^2, \quad V = \text{Var } h = \sum_{k \geq 1} \|h_k\|^2, \quad G = \mathbb{E}|\nabla h|^2 = \sum_k k \|h_k\|^2.$$

From them we get the higher-chaos share $1 - F/V$ and the mean degree of the higher chaos, $\langle k \rangle_{\geq 2} = 1 + (G - V)/(V - F)$. Hutchinson’s identities give unbiased estimates,

$$G = \mathbb{E}_{x,v}(\nabla h \cdot v)^2, \quad F = \mathbb{E}_{x,x',v}(\nabla h(x) \cdot v)(\nabla h(x') \cdot v),$$

with independent x, x' and a shared probe v . On official network 0, with 49152 inputs and totals over neurons:

layer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
F/V	.73	.60	.51	.44	.39	.36	.33	.29	.27	.26	.24	.23	.22	.21	.20	.19
$\langle k \rangle_{\geq 2}$	2.8	3.4	4.1	4.9	5.6	6.5	7.4	8.4	9.4	10.4	11.5	12.7	13.8	14.8	16.0	17.1

Layer 1 is exact for relu of a centred Gaussian: $F/V = \frac{1}{4}/(\frac{1}{2} - \frac{1}{2\pi}) = 0.733$ and $\langle k \rangle_{\geq 2} = 2.75$.

Proposition 4.1 (operator growth; measured). *As functions of the input, the neurons’ first chaos decays to under a fifth of their variance. The mean degree of the rest grows linearly in depth, about one degree per layer.*

This is the classical counterpart of operator spreading in scrambling circuits, where the size of a Heisenberg operator grows linearly in time. Its consequence for algorithm design is blunt. A degree- d chaos component in $n = 1024$ variables has about $n^d/d!$ coefficients. Any expansion in a basis fixed in advance, whether Pauli-path or Heisenberg propagation in input Hermite coordinates, would need degree around 17 at the output. It is hopeless.

4.2 Why closures work at all: frames that co-evolve with the weights

Stage 15’s wall-jet theorem expands each neuron’s relu in the Appell (Wick) polynomials of *its own layer’s law*. In that frame the births at one layer have low order: stage 14’s kink law puts

order K at resolution radius $2\sqrt{K}$, and $K \leq 4$ suffices. Composing frames telescopes the degree. Measured in the fixed input frame, the degree grows by one per layer; each frame change $\mu_l^{-1} \star \text{id}$ absorbs that growth, so in the moving frames it stays bounded.

- *The frame is a function of the instance.* The frame of layer l is (μ_l, C_l) , computed from $W_1, \dots, W_l$.
- *So the interdependence is forced.* The representation co-evolves with the sampled weights, and any tractable estimator must have this property. The chain is one instance of it, not the template.
- *Quantum reading.* Classical simulations of random circuits by Pauli propagation truncate operator weight in a fixed basis. For deep RELU networks the truncation must be done in moving frames, which is the interaction picture with the Wick map as frame change.

The open question this raises is about frame *families*. Gaussian centred frames are one choice. Section 6 shows that reshaping a Gaussian frame variationally does not help; non-Gaussian frames are untested.

5 Transport memory: holonomy, the normalizer, and lattice mismatch

5.1 The covariance chain does not determine the transport

Fluctuations, and births, are carried from z_{l-1} to z_l by the first jet $B_l = W_l \text{diag } \Phi(r_l)$. Whiten with the backbone covariances:

$$\tilde{O}_l = C_l^{-1/2} B_l C_{l-1}^{1/2}, \quad \tilde{O}_l \tilde{O}_l^\top = I - C_l^{-1/2} \Sigma_l^{\text{birth}} C_l^{-1/2}, \quad \Sigma_l^{\text{birth}} = C_l - B_l C_{l-1} B_l^\top \succeq 0.$$

So $\tilde{O}_l$ is a contraction times a rotation U_l . The contraction is the share of variance newly born from higher Mehler chaos. The rotation is the network's frame turning, and (C_{l-1}, C_l) leaves it free: every $C_l^{1/2} R C_{l-1}^{-1/2}$ with $R \in O(n)$ is compatible with the same pair of covariances. The holonomy note makes the same point sharply. Even the canonical symmetric-squeeze connection $\dot{T} = \frac{1}{2} \dot{Q} Q^{-1} T$ depends on the *path* of covariances, through the curvature $\frac{1}{4} [H Q^{-1}, K Q^{-1}]$, and a contact-preserving loop returns with a rotation (checked, Section 2).

Proposition 5.1 (transport memory; proved). *Any estimator that propagates non-Gaussian content must carry the actual transports B_l (or their action on its sources), and not only the backbone (μ_l, C_l) . Two instances with identical backbones but different rotations U_l send the same births to different places.*

5.2 The normalizer: why slices cannot be propagated as slices

Theorem 5.2 (hard-core normalizer, from the holonomy note). *An orthogonal R preserves the square-free quadratic sector $\text{span}\{z_i z_j : i < j\}$ if and only if it is a signed permutation.*

Corollary 5.3 (no closed diagonal closure; proved). *Suppose a closure keeps only neuron-diagonal data of the non-Gaussian state at each layer: the hub slices $D3$, $D21$ and κ_4 diagonals. It is closed under transport exactly when the transport, in the neuron bases, is a signed permutation. For Gaussian W_l this fails almost surely. Slices must therefore be regenerated at every readout from sources carried with their legs, or from rows transported backwards.*

This is the exact form of stage 9’s two empirical laws: Hadamard readouts defeat Frobenius truncation, and old slices correlate only 0.3–0.5 with young ones. A hub-localized leg e_m becomes, after one transport, the dense column $\Phi_m W_{:,m}$, with inverse participation ratio of order $3/n$. The chain’s CP legs are one realization of transport memory. Stage 15’s Heisenberg reads are the other.

5.3 Lattice reading

$$B_n = \text{Aut}(\mathbb{Z}^n) \cap O(n).$$

- Each layer’s neuron basis defines a coordinate lattice $\mathbb{Z}^n$ in pre-activation space. Its reflections are the folds and its orthants are the cells (stage 14: the layer map $z = Wx$ is Klartag’s normalizing map).
- The diagonal (hub) sector is the invariant sector of that lattice’s automorphism group.
- The transport between consecutive layers is a generic linear map, not an automorphism of the next layer’s lattice.

So *the memory obstruction is a lattice mismatch*: consecutive coordinate lattices are in general position. In the holonomy note’s example ($\Lambda = \mathbb{Z}e_1 \oplus 2\mathbb{Z}e_2$), contact-preserving deformations still produce rotations outside $\text{Aut}(\Lambda)$, and irrational ones produce a noncommutative torus. In the network the depth path is not a loop, but the same non-integrability makes the frame unrecoverable from its endpoints.

6 The carré du champ, Dirac energies, and two designs ruled out

6.1 The product defect of the input heat flow is the network’s covariance

For the Ornstein–Uhlenbeck flow on inputs, $\Gamma(f, g) = \nabla f \cdot \nabla g$. The companion’s accumulated product-defect identity, integrated to infinite time, gives

$$\text{Cov}(f(X), g(X)) = \int_0^1 \mathbb{E}[\nabla f(X) \cdot \nabla g(X_\rho)] d\rho, \quad X_\rho = \rho X + \sqrt{1 - \rho^2} X'.$$

For neurons this is an integral over input overlaps of two-replica Jacobian correlators, $\text{Cov}(h_l) = \int_0^1 \mathbb{E}[J_l(X) J_l(X_\rho)^\top] d\rho$. *Check* on a small He network ($n = 32$, $L = 3$): the relative Frobenius difference from the Monte Carlo covariance is 0.008, against noise of about 0.004, with diagonal ratio 0.998.

The companion’s Dirac energy, with coordinate derivations and $G = I$, is $\text{Tr } C_\rho(A) = \mathbb{E}|\nabla A|^2$, the G of Section 4. Its excess over the variance is the Poincaré deficit, $\mathbb{E}|\nabla h|^2 - \text{Var } h = \sum_k (k - 1) \|h_k\|^2 \geq 0$: the higher-chaos (magic) energy of the neuron.

6.2 Design ruled out: the leaf estimator with global Dirac energies

For a bias-free ReLU network and Gaussian input, every output is 1-homogeneous. Euler and Stein then give the exact leaf decomposition

$$\mathbb{E}f = \mathbb{E}\Delta f = \sum_{l,j} p_{z_{lj}}(0) \mathbb{E}[|\nabla z_{lj}|^2 \partial f / \partial h_{lj} \mid z_{lj} = 0],$$

the transverse measure of each wall times the energy and sensitivity on it (stage 11). This offers a second estimator, independent of any law closure. Decouple each wall term into

- the backbone wall density $\varphi(r)/s$;
- the *global* Dirac energy $(W_l K_{l-1} W_l^\top)_{jj}$, from the Jacobian-Gram recursion

$$K_l = P_l \circ (W_l K_{l-1} W_l^\top),$$

with gate second moments P_l ;

- the first-jet cocycle as the sensitivity.

It costs about $2n^3$ per layer. On network 0 it is exact at layer 1 (MSE 7×10^{-10} , the truth's noise). At layer 2 its MSE is 3.5×10^{-2} , against the Gaussian closure's 1.5×10^{-7} , and it grows to 8 at layer 16.

- *Why it fails.* The global energy exceeds the variance of z by the Poincaré deficit. The measured ratio of mean Dirac energy to variance is 1.00, 1.47, 2.54 and 13.7 at layers 1, 2, 4 and 16, which is the chaos ladder of Section 4.
- *Where the exact identity hides the cancellation.* In the exact identity this excess is cancelled between layers: a wall at layer $l - 1$ conditions the gates above it.
- *Consequence.* The leaf decomposition is exact but not term-wise stable, and the energy that enters the mean is the *transverse* one, $\mathbb{E}[|\nabla z|^2 \mid z = 0]$. In the companion's language, contact matrices must be evaluated in the leaf-conditioned state, not the global state.

This also explains why stage 9's Euler–Stein merge had to use the chain's own Laplacian response rather than a wall sum.

6.3 Design ruled out: minimal-sensitivity Wick frames

The truncated Wick expansion of $\mathbb{E}\text{relu}(Z - c)$ around a Gaussian frame $N(\kappa_1, v)$ depends on v ; the exact answer does not. Principles of minimal sensitivity pick a stationary v instead of the centred $v = \kappa_2$. For $Z = g + a(g^2 - 1)$:

case	centred frame error	stationary frames (v/κ_2 : error)
$a = 0.15$, wall 0.3, order 3	$+4.7 \times 10^{-3}$	1.22: $+2.3 \times 10^{-3}$
$a = 0.15$, wall 0.3, order 4	-1.0×10^{-2}	0.69: -9.2×10^{-3} ; 1.22: -1.1×10^{-2}
$a = 0.15$, wall -0.8 , order 4	$+1.8 \times 10^{-3}$	0.51: -1.9×10^{-2}
$a = 0.3$, wall 0.5, order 4	-2.5×10^{-2}	1.41: -2.8×10^{-2}

Stationary frames are sometimes absent and are no better on balance. The centred frame, which is EFPTZ's unique centring character, is the right choice of variance. Non-Gaussian frame *families* remain untested.

7 Interdependence of weights and estimate, made exact

The question was how an optimal system could fail to couple the sampled weights and its estimate, perhaps dynamically. There are four exact couplings, ordered by how much they are forced.

7.1 Forced: the frame

Section 4 shows that the representation must be re-centred at every layer by the instance’s own law. The frame of layer l is a function of $W_1, \dots, W_l$, and it changes as the computation proceeds.

7.2 Free: the estimator learns its own correction from its own rows

Theorem 7.1 (row-exchangeable self-calibration; proved). *Fix a layer l and the upstream state U , meaning the weights below l and the estimator’s state up to layer $l-1$. For each row w_i of W_l , let $\pi_i = \pi(w_i; U) \in \mathbb{R}^p$ be a cheap per-row feature vector and $\beta_i = \beta(w_i; U)$ an expensive per-row quantity, for instance a higher-order birth. Choose a uniformly random set S of q rows, fit $\hat{a}$ by least squares of β on π over S , and set $\hat{\beta}_i = \hat{a} \cdot \pi_i$ off S . Then:*

1. *Given U , the pairs (π_i, β_i) are i.i.d., because the rows are.*
2. *$\hat{a}$ converges to the population regression a_* , with excess error $O(p/q)$ relative to the residual.*
3. *The estimator’s added mean squared error at layer l is $(1 - \frac{q}{n})(1 - R^2) \mathbb{E}\beta^2 (1 + O(p/q))$, where R^2 is the population fit.*
4. *By the error cocycle, this error reaches the output additively, multiplied by the layer’s downstream gain.*

Proof. (1) The rows are i.i.d. and independent of U . (2) and (3) are least squares on i.i.d. samples. (4) is Theorem 3.2 of stage 15. $\square$

In this precise sense an estimator learns its own correction from its own weights. The training set is the instance’s rows, and what makes it valid is exchangeability, not any independence of weights and estimate. Corollary 3.3 of stage 15 bounds R^2 by the isotropic share whenever π holds only per-neuron or field features. A large R^2 needs anisotropic proxies, such as a short-window or first-order version of β itself.

Where it should pay, and the test. It pays where per-row costs dominate and such a proxy exists. Late-layer covariance births are the case: layers 11–16 hold 81% of the exact chain’s births, and 59% of the last-layer birth is covariance.

- The expensive quantity: the second-order covariance-birth readout of a row (the $(2, 2)$, $(3, 1)$ and κ_3 -pair diagrams), computed on $q \approx n/8$ rows.
- The proxy: the first-order readout plus the field jets.
- The prediction: the share R^2 of the channel is recovered at about an eighth of its readout cost.

7.3 Free: geometry coupled to the readout

The holonomy note’s principle $\mathcal{V}(w) = \langle f, K_w^{-1} f \rangle$ is convex in the weights w , and its optimality conditions equalize the edge energies of the Poisson solution. Our analogue is stage 15’s allocation:

- The predicted MSE is convex in the capture fractions.
- At the optimum the marginal MSE reduction per FLOP is equal across the (layer, row, diagram) choices used.
- The Poisson solution is the adjoint: the downstream gains.

The two are coupled self-consistently. The allocation depends on the adjoint, and the adjoint depends on what was computed. Two passes, or one pass with revision, solve it.

7.4 Free: task-preserving instruments

The companion’s condition $\sum_{\alpha} \mathcal{I}_{\alpha}^*(A_{\alpha}) = A$ has a classical realization.

- Branch on the sign of a chosen functional $u \cdot x$. By symmetry each branch has mass $\frac{1}{2}$.
- Run each branch with its own Wick frame. Then $\mathbb{E}f = \sum_{\alpha} P(\alpha) \mathbb{E}[f \mid \alpha]$ holds exactly.
- A branch resolves all orders of non-Gaussianity along u , at twice the cost.

Its value depends on concentration, and the residual is not concentrated. In the experimental session, making the covariance exact in the top 128 collective modes gave only 16–32%, which a branching tree could approach only with 2^{128} leaves. Instruments are the right tool for a coherent residual and the wrong one for an incoherent residual.

7.5 The design space

An estimator is a point in (frame family) $\times$ (partition blocks evaluated on the instance) $\times$ (rows) $\times$ (layers) $\times$ (instruments).

- *The standard chain.* Gaussian centred frames, pair and three-block diagrams, all rows, all layers, no instruments.
- *Forced by this stage.* Co-evolving frames and transport memory.
- *Ruled out here.* Leaf estimators with global energies, and minimal-sensitivity Gaussian frames.
- *Open, with protocols.* Non-Gaussian frame families, diagram selection by annealed two-copy variances, self-calibration over rows, and late-layer allocation.

8 Cross-validation with the experimental session

The experimental session tested stage 13 on v56 and on the truth of network 0, in §9 of its **notes/fresh-weight**. Its results read cleanly in this stage’s terms.

experimental finding	reading
gain saturates; increments uncorrelated ($V_l/4 \approx 0.0082$ flat from layer 13)	stage 12–13 prediction (saturation, not mean reversion)
the chain’s slices carry the gain shape (82–94% for κ_3 , K22, κ_4 diagonal)	the Brauer (dilation) component of each slice (Proposition 3.2)
setting slice gain amplitudes to the truth’s: +56%, +50% (predicted –10 to –30%)	the counterterms already hold the coherent sector; coherent errors add in amplitude across layers (stage 15), so local amplitude changes are amplified
final error has 0.4–3.7% of its energy along the mean	stage 15: collective share 0–3% for the exact chain; the residual is incoherent births
gain package with the old-tier ranks cut recovers only 5.3%	the lost old-tier content is three-block (stars), purely quenched; a Brauer package cannot replace it
top-32 / top-128 covariance modes exact: –12 to –16% / –16 to –32%; eigenvalues only: +9%	row theorem: a row’s covariance birth $\frac{1}{2} J_2 w^{\top} \delta C w$ weighs every mode by its share of $\ \delta C\ _F^2$; eigenvalues alone leave the anisotropic error
layers 0–6 create a few percent of the final error	stage 15 measured 5% (layers 1–6)

What this suggests for the experimental session.

1. *Leave the coherent sector alone.* It is held by the counterterms. Further work on it is fighting the amplification mechanism.
2. *Go after the late-layer covariance births, full rank.* These are the 59% channel. Use the consistent second-order diagram set, the $(2, 2)$, $(3, 1)$ and κ_3 -pair terms of the wall-jet expansion at matching order, in layers 11–16. Fund it from layers 1–6, and refit the counterterms afterwards.
3. *Decide self-calibration with one cheap diagnostic first.* At layers 12–16 on network 0, measure the R^2 of the per-row second-order covariance birth on its first-order proxy and the field jets. If $R^2 \gtrsim 0.7$, Theorem 7.1 cuts the readout cost of the channel about eightfold.

9 Checks

`scripts/verify_stage16.py` runs in about six minutes. Most of that is the Jacobian scan of network 0; set `SKIP_SLOW=1` to skip it. The transcript is `notes/stage16/verify_stage16.txt`.

statement	result
spin branching $p_{\pm}$	$(j+1)/(2j+1)$, $j/(2j+1)$, sum 1 at $j = \frac{1}{2}, 1, \frac{7}{2}$
three-spin logical qubit	$S_{12} = -Z$, $S_{23} = \frac{1}{2}Z + \frac{\sqrt{3}}{2}X$, $[S_{12}, S_{23}] = -i\sqrt{3}Y$ exactly
Zhou v1 multi-time display	$\mathbb{E}[Z_a^2 Z_b^2]$: 3.021 (MC) vs $ab + 2a^2 + a = 3.010$; product of means 1.33
isotropic angular reference G_1	0.5 and 0.0998–0.1004 (pred. 0.1), off-diagonal $\leq 3 \times 10^{-4}$
squeeze holonomy of a contact loop	angle/ $(a^2/4) = 0.985, 0.965, 0.935$ at $a = 0.02, 0.05, 0.1$; orthogonal to 5×10^{-13}
hard-core normalizer	z_i^2 leak 0 (signed permutation), 0.39 (random rotation)
mixture $\kappa_4 = \text{Cov}(Q)$	1.573/1.558, 0.538/0.518, $-0.007/-0.001$
Bernoulli realization relation	$:b^2:- :b:- = -2p(B-p)$ exactly
commutant tower	15, 4, 3.01 ($2k=4$) and 203, 31, 15.17 ($2k=6$) for $S_n, B_n, O(n)$
replica covariance identity	relative Frobenius 0.008 (noise ≈ 0.004), diagonal ratio 0.998
chaos ladder (network 0)	F/V : 0.73 $\rightarrow$ 0.19; $\langle k \rangle_{\geq 2}$: 2.76 $\rightarrow$ 17.14 over layers 1 $\rightarrow$ 16
isotropic share of truth – Gaussian	0.60–0.66; exact chain captures 0.991–0.997
minimal-sensitivity frames	no gain over the centred frame (table in Section 6)
wall-sum estimator	MSE 3.5×10^{-2} (layer 2), 8.3 (layer 16); Dirac/variance 1.47, 13.7

Not checked numerically:

- the self-calibration theorem, whose protocol is in Section 8;
- the claim that three-block content is purely quenched, which follows from Proposition 3.2(1), beyond the vanishing of annealed odd cumulants of pre-activations.